SANDIP UNIVERSITY
SCHOOL OF COMPUTER SCIENCE & ENGINEERING
Department of Computer Science and Engineering
MAHIRAVANI, TRIMBAK ROAD, TAL & DIST: NASHIK-422213, MAHARASHTRA,INDIA

B. Tech. Computer Science and Engineering (Specialization in Artificial Intelligence & Machine Learning)
Semester – V

Sr. No.	Course Type	Course Code	Course Name	Teaching Scheme (Hrs./Week)				Examination Scheme				Total Marks
				L	T	P	C	Formative Assessment CIA		Summative Assessment ESE		
								Theory	Lab	Theory	Lab	
1	PCC	NYCM501	Design and Analysis of Algorithm	3	--	--	3	50	--	100	--	100
2	PCC	NYCM502	Operating System	3	--	--	3	50	--	100	--	100
3	PCC	NYCM503	Computer Vision	3	--	--	3	50	--	100	--	100
4	PCC	NYCM511	Operating System Laboratory	--	--	4	2	--	50	--	50	100
5	PCC	NYCM512	Computer Vision Laboratory	--	--	2	1	--	25	--	25	50
6	PEC	NYCME_	Program Elective-I	3	--	--	3	50	--	100	--	100
7	OE		Open Elective III	2	--	--	2	50	--	100	--	100
8	MDM		Minor course 3 #	3	--	--	3	50	--	100	--	100
9	AEC (HSSM)	NHSA13	Essential Aptitude Skills	--	--	4	2	--	50	--	50*	100
10	VAC		Tableau and Power BI	--	--	2	--	--	--	--	--	--
TOTAL				**17**	**--**	**12**	**22**	**300**	**125**	**600**	**125**	**850**

Year: Third Year **Semester – V**
Course: Design and Analysis of Algorithm **Course Code: NYCM501**

Teaching Scheme (Hrs. /Week)				Continuous Internal Assessment (CIA)					End Semester Examination		Total
L	T	P	C	CIA-1	CIA-2	CIA-3	CIA-4	Lab	Theory	Lab	
3	0	0	3	10	20	10	10	--	50	--	100
Max. Time, End Semester Exam (Theory) - 3Hrs.									End Semester Exam (Lab) - 0Hrs.		

Prerequisites:
Data Structures And Algorithm, Basic Mathematics

Objectives:	
Students are able to:-	
1	To provide mathematical foundations for analyzing the complexity of the algorithms
2	To impart the knowledge on various design strategies that can help in solving the real world problems effectively
3	To synthesize efficient algorithms in various engineering design situations
4	To distinguish between solvable (P) and hard (NP) problems and learn to apply approximation and randomized techniques when an exact solution is computationally impossible.
5	To explore advanced frontiers like String Matching for NLP and Quantum Algorithms to understand the future of high-speed computation.

Unit No	Details	Hours
1	**Design Paradigms: Greedy, Divide and Conquer Techniques** Overview and Importance of Algorithms - Stages of algorithm development: Describing the problem, Identifying a suitable technique, Design of an algorithm, Derive Time Complexity, Proof of Correctness of the algorithm, Illustration of Design Stages - Greedy techniques: Fractional Knapsack Problem, and Huffman coding - Divide and Conquer: Maximum Subarray, Karatsuba faster integer multiplication algorithm	9
2	**Design Paradigms: Dynamic Programming, Backtracking and Branch & Bound Techniques** Dynamic programming: Assembly Line Scheduling, Matrix Chain Multiplication, Longest Common Subsequence, 0-1 Knapsack, TSP- Backtracking: N-Queens problem, Subset Sum, Graph Coloring- Branch & Bound: LIFO-BB and FIFO BB methods: Job Selection problem, 0-1 Knapsack Problem.	**9**

3	**String Matching Algorithms and Graph Algorithms** Naïve String-matching Algorithms, KMP algorithm, Rabin-Karp Algorithm, Suffix Trees. All pair shortest path: Bellman Ford Algorithm, Floyd-Warshall Algorithm - Network Flows: Flow Networks, Maximum Flows: Ford-Fulkerson, Edmond-Karp, Push Re-label Algorithm – Application of Max Flow to maximum matching problem	9
4	**Geometric Algorithms and Randomized algorithms** Line Segments: Properties, Intersection, sweeping lines - Convex Hull finding algorithms: Graham's Scan, Jarvis' March Algorithm. Randomized quick sort - The hiring problem - Finding the global Minimum Cut.	9
5	**Classes of Complexity and Approximation Algorithms** The Class P - The Class NP - Reducibility and NP-completeness – SAT (Problem Definition and statement), 3SAT, Independent Set, Clique, Approximation Algorithm – Vertex Cover, Set Cover and Travelling salesman.	9

Outcomes:	
On completion of the course, learner will be able to :-	
CO1:	Apply the mathematical tools to analyze and derive the running time of the algorithms
CO2:	Demonstrate the major algorithm design paradigms.
CO3:	Explain major graph algorithms, string matching and geometric algorithms along with their analysis.
CO4:	Articulating Randomized Algorithms.
CO5:	Explain the hardness of real-world problems with respect to algorithmic efficiency and learning to cope with it.

Text Books

1. Thomas H. Cormen, C.E. Leiserson, R L.Rivest and C. Stein, Introduction to Algorithms, Third edition, MIT Press, 2009..

Reference Books

1.Jon Kleinberg and ÉvaTardos, Algorithm Design, Pearson Education, 1st Edition, 2014.
2. Rajeev Motwani, Prabhakar Raghavan; Randomized Algorithms, Cambridge University Press, 1995 (Online Print – 2013)
3. Ravindra K. Ahuja, Thomas L. Magnanti, and James B. Orlin, Network Flows: Theory, Algorithms, and Applications, 1st Edition, Pearson Education, 2014.

Online Resources

SANDIP UNIVERSITY, NASHIK, MS (INDIA)
SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
Department of Computer Science and Engineering
MAHIRAVANI, TRIMBAK ROAD, TAL & DIST: NASHIK-422213, MAHARASHTRA,INDIA

Year: Third Year | **Semester – V**

Course: Operating System | **Course Code:** NYCM502

Teaching Scheme (Hrs. /Week)				Continuous Internal Assessment (CIA)					End Semester Examination		Total
L	T	P	C	CIA-1	CIA-2	CIA-3	CIA-4	Lab	Theory	Lab	
3	0	0	3	10	20	10	10	--	50	--	100
Max. Time, End Semester Exam (Theory) - 3Hrs.									End Semester Exam (Lab) - 0Hrs.		

Prerequisites:
Fundamentals of Computer Organization and Architecture, Data Structures, C Programming

Objectives:	
Students are able to:-	
1	To introduce the fundamental concepts, design issues, and core functionalities of operating systems..
2	To build skills in managing processes, memory, files, devices, and ensuring protection and security, using algorithms and practical simulations..
3	To introduce modern developments like virtualization, distributed operating systems, and cloud-based systems.
4	Explain the principles of file systems, storage allocation, disk scheduling, and input/output management.
5	Discuss the mechanisms for protection and security within operating systems and explore modern OS architectures through real-world case studies.

Unit No	Details	Hours
1	**Operating Systems and Structure** Module 1: Operating system basics - History of operating systems - OS structure: Monolithic, Layered, Microkernel, Modular - OS services - System calls and types - Interrupts - OS design issues – Building and booting an OS.	9
2	**Process, Scheduling, Threads and Deadlocks** Process concepts and states - Process Scheduling - Operation on Processes: Creation, Termination and Cooperation - Threads and multithreading models - Scheduling criteria - CPU scheduling: Non-preemptive (FCFS, SJF, Priority), Preemptive (Round Robin, SRTF, Preemptive Priority), Multi-level Queue, Multi level Feedback Queue - Deadlock characterization - Resource Allocation Graphs (RAG) – Deadlock handling methods. Process Synchronization: Critical Section Problem, Peterson's Solution; Hardware Solutions: Test-and-Set, Compare-and-Swap; Semaphores: Binary and Counting; Classic Synchronization Problems: Producer-Consumer,Reader-Writers, Dining Philosophers; Monitors and Condition Variables	9

3	**Process Synchronization** IPC: Shared memory, message passing - Race condition – Critical section problem - Peterson's solution – Bakery Algorithm - Hardware synchronization: Test-and-Set, Swap - Mutex locks - Semaphores – Classical synchronization problems: Bounded buffer, Readers-Writers, Dining Philosophers – Monitors.	9
4	**Memory and File Management** Address binding - Logical and physical address space - Contiguous memory allocation - Fragmentation - Placement Algorithms: First Fit, Best Fit, Worst Fit – Paging: Structure with TLB - Segmentation - Virtual memory: Demand paging - Page replacement: FIFO, Optimal, LRU – Thrashing and Working Set - File system concepts - File access methods - Directory structures and implementation - File allocation methods	9
5	**Managing Devices, Security and Protection and Virtualization** I/O device management basics: RAID structure, Disk structure, Disk Scheduling Algorithms (FCFS, SSTF, SCAN, C-SCAN, LOOK, C-LOOK), Program and network threats – Cryptography as a security tool – Domains of protection – Access matrix – Capability based systems - Need for virtualization - Virtual machines and architectures – Hypervisors - Virtualization Technologies: Para Virtualization, Full Virtualization - OS-Level Virtualization: Containers, Docker Basics, Containers vs Virtual Machines.	9

Outcomes:	
On completion of the course, learner will be able to :-	
CO1:	Understand the structure, functionalities, and foundational concepts of operating systems.
CO2:	Apply process scheduling algorithms and analyze thread models and deadlock-handling methods
CO3:	Implement synchronization mechanisms and solve concurrency problems in multi-process systems.
CO4:	Evaluate memory management and file system strategies used in operating systems
CO5:	Analyze device management, protection mechanisms, and virtualization techniques in operating systems.

Text Books

1 . Abraham Silberschatz, Peter Baer Galvin, Greg Gagne, Operating System Concepts (Dinosaur Book), Wiley, 10th Edition.

Reference Books

1. Remzi H. Arpaci-Dusseau, Andrea C. Arpaci-Dusseau, "Operating Systems: Three Easy Pieces", Arpaci Dusseau Books, 1st Edition, 2023.
2. Andrew S. Tanenbaum, Herbert Bos, "Modern Operating Systems", Pearson Education, 4th Edition, 2015.
3. William Stallings, "Operating Systems: Internals and Design Principles", Pearson, 9th Edition, 2018.

Online Resources
https://nptel.ac.in/courses/106108101 - https://nptel.ac.in/courses/106105083 - https://www.os-book.com/OS10/ - https://www.coursera.org/learn/os-pku

Year: Third Year **Semester – V**
Course: Computer Vision **Course Code: NYCM503**

Teaching Scheme (Hrs. /Week)				Continuous Internal Assessment (CIA)					End Semester Examination		Total
L	T	P	C	CIA-1	CIA-2	CIA-3	CIA-4	Lab	Theory	Lab	
3	0	0	3	10	20	10	10	--	50	--	100
Max. Time, End Semester Exam (Theory) - 3Hrs.									End Semester Exam (Lab) - 0Hrs.		

Prerequisites:
Linear Algebra, Probability and Statistics, Data Structures and Algorithms, Basic Python Programming

Objectives:	
Students are able to:-	
1	Understand fundamentals of image formation and human vision.
2	Analyze digital image processing techniques used in computer vision.
3	Apply feature detection and extraction algorithms.
4	Understand motion analysis and object recognition techniques.
5	Relate computer vision concepts with real-world AI applications.

Unit No	Details	Hours
1	**Introduction to Computer Vision** Overview of Computer Vision, Human Visual System, Image Formation and Camera Models, Pinhole Camera Model, Image Sensors and Acquisition.	9
2	**Image Processing Fundamentals** Digital Image Representation, Point Processing Operations, Spatial Filtering, Edge Detection Techniques, Image Enhancement.	9
3	**Feature Detection and Description** Corner Detection (Harris, Shi-Tomasi), Blob Detection, Feature Descriptors (SIFT, SURF, ORB), Feature Matching Techniques.	9
4	**Motion and Depth Analysis** Optical Flow, Stereo Vision, Epipolar Geometry, Depth Estimation, Motion Tracking.	9
5	**Object Recognition and Applications** Object Detection Techniques, Template Matching, Face Detection and Recognition, Deep Learning for Computer Vision (Overview), Applications of Computer Vision.	9

Outcomes:	
On completion of the course, learner will be able to :-	
CO1:	Explain fundamentals of computer vision and image formation.
CO2:	Apply image processing techniques for enhancement and analysis.
CO3:	Implement feature detection and matching algorithms.
CO4:	Analyze motion and depth from image sequences.
CO5:	Design basic object recognition systems.

Text Books

1. R. Szeliski, *Computer Vision: Algorithms and Applications*, 2nd ed. Cham, Switzerland: Springer, 2022.
2. R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, 4th ed. New York, NY, USA: Pearson, 2018.
3. D. A. Forsyth and J. Ponce, *Computer Vision: A Modern Approach*, 2nd ed. Upper Saddle River, NJ, USA: Pearson, 2012.

Reference Books

1. M. Sonka, V. Hlavac, and R. Boyle, Image Processing, Analysis, and Machine Vision, 4th ed. Boston, MA, USA: Cengage Learning, 2014.
2. E. Trucco and A. Verri, Introductory Techniques for 3D Computer Vision, Upper Saddle River, NJ, USA: Pearson, 1998.

Online Resources

- NPTEL: Computer Vision Courses
- OpenCV Documentation
- Coursera & edX Computer Vision Courses

Year: Third Year **Semester –V**

Course: Operating System Laboratory

Course Code: NYCM511

Teaching Scheme (Hrs. /Week)				Continuous Internal Assessment (CIA)					End Semester Examination		Total
L	T	P	C	CIA-1	CIA-2	CIA-3	CIA-4	Lab	Theory	Lab	
3	0	0	2	--	--	--	--	10	--	50	50
Max. Time, End Semester Exam (Theory) - 0Hrs.									End Semester Exam (Lab) - 2Hrs.		

Prerequisites:
Basics of Operating Systems, Introduction to the Main Functions of an Operating System, and Operating System Processes.

Objectives:	
Students are able to:-	
1	Understand the concept of operating systems, its principles, and implement basic shell scripting and system calls.
2	Apply various CPU scheduling algorithms and thread management techniques to optimize processor utilization.
3	Implement synchronization primitives and deadlock handling mechanisms to resolve concurrency issues.
4	Analyze and simulate memory management strategies including allocation, paging, and segmentation.
5	Design and evaluate I/O management policies, disk scheduling algorithms, and file system organizations.

Unit No	Details	Hours
1	Study of basic Linux commands and shell scripting	2
2	Implementation of Bootloader	2
3	Process creation using fork(), Orphan and Zombie processes	2
4	Thread Creation	2
5	CPU scheduling algorithms (FCFS, SJF, Priority)	2
6	CPU scheduling algorithms (Round Robin, SRTF, Pre-emptive Priority)	2
7	Banker's Algorithm for Deadlock Avoidance (Safe state checking and additional resource request granting)	2
8	Shared Memory Creation and sharing contents between processes	2
9	Implementation of Producer-Consumer/Readers Writers using Semaphore	2
10	Implementation of Dining Philosophers using Semaphore	2
11	Memory Allocation Strategies: First Fit, Best Fit, Worst Fit	2

12	Page Replacement Algorithms: FIFO, LRU, Optimal	2
13	Disk Scheduling Algorithms: FCFS, SSTF, SCAN	2
14	Disk Scheduling Algorithms: C-SCAN, LOOK, C-LOOK	2
15	File system operations using system calls	2

Outcomes:	
On completion of the course, learner will be able to :-	
CO1:	Understand the concept of operating systems, its principles, and implement basic shell scripting and system calls.
CO2:	Apply various CPU scheduling algorithms and thread management techniques to optimize processor utilization.
CO3:	Implement synchronization primitives and deadlock handling mechanisms to resolve concurrency issues.
CO4:	Analyze and simulate memory management strategies including allocation, paging, and segmentation.
CO5:	Design and evaluate I/O management policies, disk scheduling algorithms, and file system organizations.

Text Books

1. Abraham Silberschatz, Peter Baer Galvin, Greg Gagne., Operating System Concepts., Wiley; 10th Edition.
2. Andrew S. Tanenbaum, Herbert Bos., Modern Operating Systems., Pearson; 4th Edition.

Reference Books

1. William Stallings., Operating Systems: Internals and Design Principles., Pearson; 9th Edition.
2. Sumitabha Das., UNIX Concepts and Applications., Tata McGraw Hill; 4th Edition.

Online Resources : E-Learning Resources:

- https://onlinecourses.nptel.ac.in/noc22_cs91/preview
- https://www.coursera.org/learn/bits-operating-systems
- https://www.tutorialspoint.com/operating_system/index.htm
- https://www.udemy.com/topic/operating-system/

Year: Third Year **Semester –V**
Course: Computer Vision Laboratory **Course Code: NYCM512**

Teaching Scheme (Hrs. /Week)				Continuous Internal Assessment (CIA)					End Semester Examination		Total
L	T	P	C	CIA-1	CIA-2	CIA-3	CIA-4	Lab	Theory	Lab	
0	0	2	1	--	--	--	--	25	--	25	50
Max. Time, End Semester Exam (Theory) - .									End Semester Exam (Lab) – 02 Hrs.		

Prerequisites:
Linear Algebra, Probability and Statistics, Data Structures and Algorithms, Basic Python Programming

Objectives:	
Students are able to:-	
1	Practice basic image manipulation and geometric transformations.
2	Implement spatial filtering techniques for noise reduction and sharpening.
3	Apply edge and feature detection algorithms to real-world images.
4	Execute image segmentation and object detection scripts.
5	Develop end-to-end vision applications using deep learning frameworks.

Experiment List: -The student shall complete the following experiments.		
Expt. No. 1	Basic image operations: Reading, displaying, and geometric scaling/rotation.	CO1
Expt. No. 2	Implementation of image enhancement using Histogram Equalization.	CO1
Expt. No. 3	Application of Smoothing (Blurring) and Sharpening spatial filters.|	CO2
Expt. No. 4	Edge Detection implementation using Canny and Sobel operators.	CO2
Expt. No. 5	Keypoint detection and description using SIFT or SURF.	CO3
Expt. No. 6	Image Stitching/Alignment based on feature matching.	CO3
Expt. No. 7	Image Segmentation using K-Means clustering or Thresholding.	CO4
Expt. No. 8	Object Detection using Haar Cascade Classifiers (Face/Eye detection).	CO4
Expt. No. 9	Image Classification using a pre-trained CNN (e.g., MobileNet).	CO5
Expt. No. 10	Real-time Video Analysis: Motion detection or Background subtraction.	CO5

Outcomes:	
At the end of this course students will be able to	
CO1	Perform fundamental image manipulation and preprocessing.
CO2	Design filters to improve image quality and highlight features.
CO3	Implement robust feature matching between different views of a scene.
CO4	Extract meaningful objects from images through segmentation.
CO5	Create functional computer vision applications for practical problems.

Text Books

1. G. Bradski and A. Kaehler, Learning OpenCV 4: Computer Vision with Python, 3rd ed. O'Reilly Media, 2019.
2. R. Laganiere, OpenCV 4 Computer Vision Application Programming Cookbook, 4th ed. Packt Publishing, 2019.
3. A. Geron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems, 3rd ed. O'Reilly Media, 2022.

Reference Books

1. M. Nixon and A. Aguado, Feature Extraction & Image Processing for Computer Vision, 4th ed. Academic Press, 2019.
2. S. Khan, H. Rahmani, S. A. A. Shah, and M. Bennamoun, A Guide to Convolutional Neural Networks for Computer Vision. Morgan & Claypool Publishers, 2018.
3. R. Szeliski, Computer Vision: Algorithms and Applications, 2nd ed. Springer, 2022.
4. J. Howse and J. Minichino, Learning OpenCV 4 Computer Vision with Python 3, 3rd ed. Packt Publishing, 2020.
5. D. A. Forsyth and J. Ponce, Computer Vision: A Modern Approach, 2nd ed. Pearson, 2011.

Online Resources : E-Learning Resources:

- **NPTEL / SWAYAM:**
  - *Computer Vision* by Prof. Jayanta Mukhopadhyay (IIT Kharagpur).
  - *Introduction to Machine Vision* by Prof. G.N. Rathna (IISc Bangalore).
- **Coursera:**
  - *Introduction to Computer Vision* (provided by Georgia Tech).
  - *Deep Learning Specialization* (DeepLearning.AI) – specifically the "Convolutional Neural Networks" course.
- **EdX:**
  - *Computer Vision and Image Analysis* (provided by Microsoft).
- **Open Source Libraries & Documentation:**
  - **OpenCV Official Documentation:** Comprehensive tutorials for C++ and Python (https://docs.opencv.org/).
  - **PyTorch/TensorFlow Vision Modules:** Documentation for torchvision and tf.keras.preprocessing.
- **Interactive Learning Platforms:**
  - **Kaggle:** Computer Vision micro-courses and hands-on competitions for image classification and segmentation.
  - **PyImageSearch:** Practical tutorials on OpenCV, Deep Learning, and Computer Vision by Adrian Rosebrock.

Year: Third Year **Semester – V**

Course: Theory of Computation **Course Code: NYCME01**

Teaching Scheme (Hrs. /Week)				Continuous Internal Assessment (CIA)					End Semester Examination		Total
L	T	P	C	CIA-1	CIA-2	CIA-3	CIA-4	Lab	Theory	Lab	
3	0	0	3	10	20	10	10	--	50	--	100
Max. Time, End Semester Exam (Theory) - 3Hrs.									End Semester Exam (Lab) - 0Hrs.		

Prerequisites:
Linear Algebra, Probability and Statistics, Data Structures and Algorithms, Basic Python Programming

Objectives:	
Students are able to:-	
1	To introduce the fundamental concepts of formal languages, grammars, and automata theory to provide a foundation for the theoretical understanding of computation.
2	To explore the design and analysis of finite automata, regular expressions, context-free grammar, pushdown automata, and Turing machines, emphasizing their role in language recognition and parsing
3	To analyze problems related to decidability, reducibility, and undecidability, and to understand the limitations of algorithmic computation
4	Analyze computational models such as Turing machines and their role in defining computability.
5	Evaluate decidability, undecidability, and complexity concepts in formal language theory

Unit No	Details	Hours
1	**Foundations of Formal Languages and Proof Techniques:** Proof techniques: Induction, Contradiction, Contrapositive, Introduction to formal languages and operations - Types of grammars - Chomsky hierarchy- Overview of models of automata	9
2	**Finite Automata and Regular Languages:** Context-Free Grammars (CFG) - Context-Free Languages - Derivations – Ambiguous and Unambiguous grammars - First and Follow - Left Recursion and Left Factoring – Simplification of CFG: Chomsky Normal Form – Greibach Normal Form - CYK Algorithm	9
3	**Context Free Grammars and Languages:** Definition and types of grammar, Chomsky hierarchy of grammar, Relation between types of grammars, Context free grammar, Left most & right most derivation trees, Ambiguity in grammar, Simplification of context	9

	free grammar, Chomsky Normal From, Greibach Normal From, properties of context free language, Pumping lemma for context free language, Decision algorithm for context tree language, CYK parsing algorithm	
4	**Parsing and Pushdown Automata:** Introduction to parsing: Top-down and Bottom-up - LL (1) Parsing - SLR Parsing Pushdown Automata (PDA) - Deterministic and Non-deterministic PDA - 11 hours Interpretation of syntactic statements using PDA.	9
5	**Turing Machines and Undecidability:** Introduction and Basic Turing Machines (acceptors and transducers) - Universal Turing machine and encoding - Church Turing Thesis - Recursive Enumerable Languages -Recursively Enumerable Languages -Non Recursive Languages -Reducibility - Undecidability: Halting Problem - Post Correspondence Problem – Tools: JFLAP, REGEX, LEX, YACC	9

Outcomes:

On completion of the course, learner will be able to :-

CO1:	Apply various proof techniques such as induction, contradiction, and contraposition in the context of formal languages and automata
CO2:	Design and analyze finite automata and regular expressions to solve lexical analysis problems and recognize regular languages
CO3:	Construct and simplify context-free grammar and apply parsing algorithms to determine syntactic structure in languages
CO4:	Develop pushdown automata and understand parsing strategies such as LL(1) and LR for context-free languages
CO5:	Understand and evaluate the computational power and limitations of Turing machines, and distinguish between decidable, semi-decidable, and undecidable problems

Text Books

1. John E. Hopcroft, Rajeev Motwani, Jeffrey D. Ullman, "Introduction to Automata

Reference Books

1. Peter Linz, Susan H. Rodger, Jones, "An Introduction to Formal Languages and Automata", Bartlett Publishers, 7th Edition, 2022
2. Dan A. Simovici, "Introduction to the Theory of Formal Languages", World Scientific, 2024
3. Alfred V. Aho, Monica S. Lam, Ravi Sethi, Jeffrey D. Ullman, Sorav Bansal, Compilers: Principles, Techniques, and Tools, Updated", Pearson India, 2nd Edition, 2006

Online Resources

- NPTEL: Computer Vision Courses
- OpenCV Documentation
- Coursera & edX Computer Vision Courses

Year: Third Year **Semester – V**
Course: Machine Learning **Course Code: NYCME02**

Teaching Scheme (Hrs. /Week)				Continuous Internal Assessment (CIA)					End Semester Examination		Total
L	T	P	C	CIA-1	CIA-2	CIA-3	CIA-4	Lab	Theory	Lab	
3	0	0	3	10	20	10	10	--	50	--	100
Max. Time, End Semester Exam (Theory) - 3Hrs.									End Semester Exam (Lab) - 0Hrs.		

Prerequisites:
Basic database concepts and data handling, basic programming or analytical thinking skills.

Objectives:	
Students are able to:-	
1	To introduce core paradigms and theoretical principles of machine learning.
2	To develop the skills to design, implement and evaluate machine learning models for data driven applications.
3	To build expertise to optimize and fine-tune machine learning systems for efficiency and generalization
4	Use Tableau to connect data sources and create interactive visualizations and dashboards for data exploration.
5	Develop KPIs, analytical calculations, and interactive dashboards to support data-driven decision making.

Unit No	Details	Hours
1	. **Introduction** Introduction - Learning Paradigms - Information Theoretic Learning - Version Spaces - Finite and Infinite Hypothesis Spaces - PAC Learning - VC Dimension - Inductive Bias - Occam's Razor - Empirical Risk Minimization - Structural Risk Minimization.	9
2	**Supervised Learning** Regression - Linear, Multiple Linear - Model Calibration and Uncertainty Estimation - Classification - Logistic Regression, KNN, Naive Bayes, Decision Tree - ID3, CART - Neural Network - Multi-Layer Perceptron - Gradient-Based Methods - Stochastic Gradient Descent, Momentum, Nesterov, Adam, RMSProp - Support Vector Machine(SVM) - Linear and Nonlinear - Loss Functions and Regularization - LASSO, RIDGE, Elastic Net, Huber - Evaluation Metrics.	9

3	**Unsupervised Learning** Clustering - Partitional - K-Means, K-Mode - Hierarchical - AGNES, DIANA - Density Based - DBSCAN - Model-Based - Expectation Maximization, Gaussian Mixture Models, Self-Organizing Map - Principal Component Analysis (PCA) - Contrastive Self Supervised Learning - SimCLR, BYOL, Manifold Learning and Spectral Clustering - Evaluation Metrics- Silhouette, Dunn, Davies, Bouldin.	9
4	.**Ensemble Learning** Ensemble Learning - Bagging - Random Forests - Boosting- AdaBoost, XGBoost, LightGBM - Calibration and Confidence Estimation in Large Models - Expected Calibration Error and Maximum Calibration Error - Bias and Variance Trade-off.	9
5	**Hyperparameter Tuning and Optimization** Convex and Non-Convex Optimization - Optimality Conditions - Unconstrained and Constrained Optimization - Optimization Variants - Trust Region Methods - Stochastic Approximation - Mini-batch Stability - Hyperparameter Tuning - Grid Search, Random Search, Bayesian Optimization.	9

Outcomes:	
On completion of the course, learner will be able to :-	
CO1:	Understand and apply various learning paradigms and principles to evaluate and design machine learning models.
CO2:	Implement supervised learning models for real-world applications and evaluate their performance using suitable metrics.
CO3:	Analyze unknown data patterns using appropriate clustering techniques
CO4:	Apply ensemble learning techniques by combining base learners to address real-time problems effectively.
CO5:	Optimize machine learning models through hyperparameter tuning and performance improvement methods.

Text Books

1. Ethem Alpaydin, "Introduction to Machine Learning", MIT Press, 4th Edition, 2020
2. AurÃ©lien GÃ©ron , "Hands-On Machine Learning with Scikit - Learn, Keras, and TensorFlow", O Reilly Media, 3rd Edition, 2023
3. Singiresu S. Rao, "Engineering Optimization: Theory and Practice", John Wiley and Sons, 4th Edition, 202

Reference Books
1. Tom M. Mitchell, "Machine Learning", McGraw Hill, 3rd Edition, 1997 2. Jiawei Han, Jian Pei, and Hanghang Tong, "Data Mining: Concepts and Techniques ", Morgan Kaufmann, 4th Edition, 2022 3. Amir Beck. , "First-Order Methods in Optimization", MOS SIAM Series on Optimization , 2nd Edition,
Online Resources
• https://nptel.ac.in/courses/117105101 • https://nptel.ac.in/courses/106106046

Year: Third Year **Semester – V**
Course: Operating System **Course Code: NYCMM04**

Teaching Scheme (Hrs. /Week)				Continuous Internal Assessment (CIA)					End Semester Examination		Total
L	T	P	C	CIA-1	CIA-2	CIA-3	CIA-4	Lab	Theory	Lab	
3	0	0	3	10	20	10	10	--	50	--	100
Max. Time, End Semester Exam (Theory) - 3Hrs.									End Semester Exam (Lab) - 0Hrs.		

Prerequisites:
Fundamentals of Computer Systems and Computer Organization, Memory and Input/output device concepts

Objectives:	
Students are able to:-	
1	To introduce the fundamental concepts, design issues, and core functionalities of operating systems..
2	To build skills in managing processes, memory, files, devices, and ensuring protection and security, using algorithms and practical simulations..
3	To introduce modern developments like virtualization, distributed operating systems, and cloud-based systems.
4	Explain the principles of file systems, storage allocation, disk scheduling, and input/output management.
5	Discuss the mechanisms for protection and security within operating systems and explore modern OS architectures through real-world case studies.

Unit No	Details	Hours
1	**Operating Systems and Structure** Module 1: Operating system basics - History of operating systems - OS structure: Monolithic, Layered, Microkernel, Modular - OS services - System calls and types - Interrupts - OS design issues – Building and booting an OS.	9
2	**Process, Scheduling, Threads and Deadlocks** Process concepts and states - Process Scheduling - Operation on Processes: Creation, Termination and Cooperation - Threads and multithreading models - Scheduling criteria - CPU scheduling: Non-preemptive (FCFS, SJF, Priority), Preemptive (Round Robin, SRTF, Preemptive Priority), Multi-level Queue, Multi level Feedback Queue - Deadlock characterization - Resource Allocation Graphs (RAG) – Deadlock handling methods. Process Synchronization: Critical Section Problem, Peterson's Solution; Hardware Solutions: Test-and-Set, Compare-and-Swap; Semaphores: Binary and Counting; Classic Synchronization Problems: Producer-Consumer,Reader-Writers, Dining Philosophers; Monitors and Condition Variables	9
	Process Synchronization IPC: Shared memory, message passing - Race condition – Critical section	

3	problem - Peterson's solution – Bakery Algorithm - Hardware synchronization: Test-and-Set, Swap - Mutex locks - Semaphores – Classical synchronization problems: Bounded buffer, Readers-Writers, Dining Philosophers – Monitors.	9
4	**Memory and File Management** Address binding - Logical and physical address space - Contiguous memory allocation - Fragmentation - Placement Algorithms: First Fit, Best Fit, Worst Fit – Paging: Structure with TLB - Segmentation - Virtual memory: Demand paging - Page replacement: FIFO, Optimal, LRU – Thrashing and Working Set - File system concepts - File access methods - Directory structures and implementation - File allocation methods.	9
5	**Managing Devices, Security and Protection and Virtualization** I/O device management basics: RAID structure, Disk structure, Disk Scheduling Algorithms (FCFS, SSTF, SCAN, C-SCAN, LOOK, C-LOOK), Program and network threats – Cryptography as a security tool – Domains of protection – Access matrix – Capability based systems - Need for virtualization - Virtual machines and architectures – Hypervisors - Virtualization Technologies: Para Virtualization, Full Virtualization - OS-Level Virtualization: Containers, Docker Basics, Containers vs Virtual Machines.	9

Outcomes:	
On completion of the course, learner will be able to :-	
CO1:	Apply the mathematical tools to analyze and derive the running time of the algorithms
CO2:	Demonstrate the major algorithm design paradigms.
CO3:	Explain major graph algorithms, string matching and geometric algorithms along with their analysis.
CO4:	Articulating Randomized Algorithms.
CO5:	Explain the hardness of real-world problems with respect to algorithmic efficiency and learning to cope with it.

Text Books

1. Abraham Silberschatz, Peter B. Galvin, Greg Gagne, “Operating System Concepts”, Publisher: Wiley India, ISBN: 978-81-265-2987-0

Reference Books

1. Remzi H. Arpaci-Dusseau, Andrea C. Arpaci-Dusseau, "Operating Systems: Three Easy Pieces", Arpaci Dusseau Books, 1st Edition, 2023
2. Andrew S. Tanenbaum, “Modern Operating Systems”, Pearson Education, ISBN: 978-93-325-5495-6
3. William Stallings, “Operating Systems: Internals and Design Principles”, Pearson Education, ISBN: 978-93-325-4532-

Online Resources

- https://nptel.ac.in/courses/106106144?utm_source=chatgpt.com
- https://learn.microsoft.com/en-us/windows/?utm_source=chatgpt.com

Year: Third Year **Semester –V**

Course: Cyber Law & Ethics **Course Code: NYCMO05**

Teaching Scheme (Hrs. /Week)				Continuous Internal Assessment (CIA)					End Semester Examination		Total
L	T	P	C	CIA-1	CIA-2	CIA-3	CIA-4	Lab	Theory	Lab	
3	0	0	2	10	20	10	10	--	50	--	100
Max. Time, End Semester Exam (Theory) - 3Hrs.									End Semester Exam (Lab) - 0Hrs.		

Prerequisites:
Basics of Computer Networks, Introduction to Information Technology, Fundamentals of Computer Science

Objectives:	
Students are able to:-	
1	To understand the legal framework governing cyberspace including the IT Act 2000 and its amendments.
2	To acquire knowledge of various types of cybercrimes, their impact, and the mechanisms for prevention and investigation.
3	To analyze ethical issues arising in the digital environment including privacy, intellectual property rights, and digital identity.
4	To explore international cyber law frameworks, treaties, and cross-border legal challenges in cyberspace.
5	To comprehend compliance requirements, data protection laws, and ethical responsibilities of IT professionals.

Unit No	Details	Hours
1	**Introduction to Cyber Law and IT Act 2000** **Module 1:** Introduction to Cyberspace and Cyber Law, Need for Cyber Legislation, Overview of Information Technology Act 2000 (IT Act 2000) and Amendment Act 2008, Key Definitions under IT Act: Electronic Record, Digital Signature, Certifying Authority. **Module 2.** Legal Recognition of Electronic Records and Digital Signatures, Electronic Governance, Regulation of Certifying Authorities, Overview of IT (Amendment) Act 2008: Key Changes and New Provisions, Controller of Certifying Authorities (CCA), Scope and Limitations of IT Act 2000.	6
2	**Cybercrimes and Legal Remedies** **Module 1:** Definition and Classification of Cybercrimes, Types of Cybercrimes: Hacking, Phishing, Identity Theft, Cyberstalking, Cyber Terrorism, Online Fraud, Child Pornography, Ransomware, Offences and Penalties under IT Act 2000 (Sections 43, 65, 66, 66A-F, 67, 72, 84B), Indian Penal Code (IPC) Provisions related to Cybercrimes.	6

	Module 2: Cyber Forensics: Concepts and Digital Evidence, Investigation and Adjudication of Cybercrimes, Role of Cyber Cells and Law Enforcement Agencies, Jurisdiction Issues in Cybercrime Cases, Case Studies on Cybercrime in India.	
3	**Intellectual Property Rights in Cyberspace** **Module 1:** Introduction to Intellectual Property Rights (IPR), Copyright Law and Digital Works: Software, Databases, Multimedia, Copyright Infringement Online and Digital Piracy, Trademarks in Cyberspace: Domain Name Disputes, Cybersquatting, Patents for Software and Business Methods. **Module 2:** Trade Secrets and Confidential Information in Digital Age, Digital Rights Management (DRM) Technologies, Creative Commons Licensing, Open Source Software Licensing: GPL, MIT, Apache and WIPO Initiatives for Cyber IPR Protection, Case Studies on Domain Name Disputes and Software Piracy.	6
4	**Privacy, Data Protection and E-Commerce Law** **Module 1:** Right to Privacy in the Digital Age, Personal Data and Sensitive Personal Data, Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Data or Information) Rules 2011, Personal Data Protection Bill (PDPB) India, General Data Protection Regulation (GDPR): Principles, Rights of Data Subjects, Obligations of Controllers and Processors. **Module 2:** E-Commerce Legal Framework: Formation of Online Contracts, Electronic Payment Systems, Consumer Protection in E-Commerce, Cybersecurity Policies and Compliance Frameworks, Role of CERT-In, Breach Notification Requirements, Social Media Regulations and Intermediary Liability.	6
5	**Ethics in Information Technology and International Cyber Law** **Module 1:** Professional Ethics for IT Practitioners, Code of Conduct and Ethics for IT Professionals, ACM and IEEE Code of Ethics, Ethical Issues in Artificial Intelligence and Big Data, Net Neutrality: Concept, Debate, and Policy, Digital Divide and Ethical Responsibilities. **Module 2:** International Cyber Law Frameworks: Budapest Convention on Cybercrime, Cross-Border Cybercrime and Jurisdiction Challenges, United Nations Initiatives on Cybersecurity, Cybersecurity Policy in India: National Cyber Security Policy 2013, Responsible Disclosure and Bug Bounty Ethics, Future Trends in Cyber Law and Digital Ethics.	6

Outcomes:	
On completion of the course, learner will be able to :-	
CO1:	Use the syntax and semantics of java programming language and basic concepts of OOP.
CO2:	Understand the features of Java such as operators, classes, objects, inheritance.
CO3:	Develop reusable programs using the concepts of inheritance, polymorphism, Interfaces and packages. Apply the concepts of Multithreading and Exception handling to develop efficient and error free codes.
CO4:	Acquire competence in Java through the use of swing, applets.
CO5:	Design event driven GUI and web related applications which mimic the real word scenarios.
Text Books	

1. Pawan Duggal, Cyber Law: The Indian Perspective, Saakshar Law Publications.

2. Vivek Sood, Cyber Law Simplified, Tata McGraw-Hill Education.

Reference Books

1. Farooq Ahmad, Cyber Law in India (Law on Internet), New Era Law Publication.
2. Nandan Kamath, Law Relating to Computers, Internet and E-Commerce, Universal Law Publishing.
3. Chris Reed, Internet Law: Text and Materials, Cambridge University Press.
4. David Bainbridge, Introduction to Information Technology Law, Pearson Education.
5. Michael L. Rustad, Global Internet Law in a Nutshell, West Academic Publishing.

Online Resources : E-Learning Resources:

- https://nptel.ac.in/courses/109104050
- https://www.meity.gov.in/content/information-technology-act
- https://www.cert-in.org.in/
- https://gdpr.eu/

Year: Third Year **Semester – 5**
Course: Introduction to Reverse Engineering (Open Elective) **Course Code: NYCMO06**

Teaching Scheme (Hrs. /Week)				Continuous Internal Assessment (CIA)					End Semester Examination		Total
L	T	P	C	CIA-1	CIA-2	CIA-3	CIA-4	Lab	Theory	Lab	
2	0	0	2	10	20	10	10	--	50	--	100
Max. Time, End Semester Exam (Theory) - 3Hrs.									End Semester Exam (Lab) - 0Hrs.		

Prerequisites:
C Programming, C++ Programming, Programming for Problem Solving, and Computer Architecture/Organization.

Objectives:	
Students are able to:-	
1	To establish a foundational understanding of reverse engineering principles, the IA-32 architecture, and the essential elements of the Windows operating system that are pertinent to binary analysis.
2	To equip students with the knowledge of essential reversing tools and demonstrate their application in analyzing undocumented software interfaces for interoperability.
3	To introduce students to the techniques of data reverse engineering for deciphering proprietary file formats and to develop skills in auditing binaries for common software vulnerabilities.
4	To provide a comprehensive understanding of various malware types and their evasion techniques, while also introducing fundamental concepts of cracking and anti-reversing.
5	To explore the unique challenges and opportunities in reversing .NET applications, including the role of obfuscators and decompilers, and to provide an overview of the frontier of native code decompilation.

Unit No	Details	Hours
1	**Foundations of Reverse Engineering and Low-Level Systems** **Module 1: Core Concepts** What is Reverse Engineering? , The Reversing Process and Toolkit , High-Level vs. Low-Level Software Introduction to the IA-32 Architecture, Key IA-32 Instructions and Control Flow **Module 2: Compilers, Execution Environments, and Windows Fundamentals** Compilers and Their Impact on Reversing, Execution Environments, Windows Fundamentals - Memory Management Windows Fundamentals - Processes, Threads, and APIs	9

2	**The Reversing Toolkit and Achieving Interoperability** **Module 1: Essential Reversing Tools** Disassemblers in, User-Mode Debugging with OllyDbg, Kernel-Mode, Debugging Concepts, System-Monitoring Tools, Auxiliary Tools **Module 2: Case Study - Reversing an Undocumented API** The Goal of Interoperability, Case Study Setup: The Generic Table API , Dissecting Data Structures and Initialization, Reconstructing Logic and Usage	9
3	**Data Reverse Engineering and Vulnerability Analysis Module 1: Deciphering Proprietary File Formats** The Art of Data Reverse Engineering, Case Study: The "Cryptex" Encrypted, Archive Tracing the Password Verification Logic, Decrypting the Directory Structure, constructing the File Extraction Algorithm **Module 2: Auditing Binaries for Vulnerabilities** The Problem of Vulnerable Stack Overflows in Depth, Integer Overflows and Type Conversion Errors Real-World Case Study: IIS Indexing Service Vulnerability	9
4	**Malware Analysis and Cracking Fundamentals Module 1: Reversing Malicious Code** A Taxonomy of Malware, Malware Defenses: Polymorphism and Metamorphism, Establishing a Safe Reversing Environment , Case Study: Unpacking and Initial Analysis of a Backdoor , Case Study: Dissecting Backdoor Functionality **Module 2: Antireversing and Cracking Fundamentals** The Purpose of Antireversing , Active Antidebugger Techniques, Introduction to Cracking Basic Cracking in Practice	**9**
5	**Advanced Topics: .NET Reversing and Decompilation** **Module 1: The .NET Reversing Landscape** Introduction to the .NET Platform, The Intermediate Language (IL) and Metadata, Reading and Analyzing IL Code , Decompilers , The Obfuscator's Response **Module 2: The Decompilation Frontier and Conclusion** The Dream of Native Code Decompilation Decompiler Architecture - The Front End Decompiler Architecture - Code Analysis and the Back End Course Wrap-up and Future Directions	**9**

Outcomes:	
On completion of the course, learner will be able to :-	
CO1:	Explain the core concepts of reverse engineering, identify key IA-32 instructions, and describe the fundamental components of the Windows OS relevant to program execution.
CO2:	Select and utilize appropriate disassemblers, debuggers, and system monitors to perform basic static and dynamic analysis of an executable, demonstrating the ability to reverse engineer an undocumented API.
CO3:	Apply a methodical approach to reverse engineer an undocumented file format and identify potential security vulnerabilities such as stack and integer overflows within a binary.
CO4:	Classify different types of malware, analyze a simple backdoor in a controlled environment, and explain basic antireversing techniques and cracking methods like patching and keygenning.
CO5:	Analyze simple .NET IL code, explain the purpose of decompilers and obfuscators in the .NET ecosystem, and summarize the fundamental challenges and techniques involved in native code decompilation.

Text Books

1. Eilam, E. (2005). *Reversing: Secrets of Reverse Engineering*. Wiley Publishing, Inc.

Reference Books

1. Bruce Dang, Alexandre Gazet, Elias Bachaalany and Sebastien Josse, Practical Reverse Engineering, First Edition, Wiley Publishers, 2014.
2. EldadEilam, Reversing: Secrets of Reverse Engineering, Wiley Publishers, 2005.
3. Chris Eagle, IDA Pro Book, Second Edition, No Starch Press, 2011.
4. Michael Sikorski and Andrew Honig, Practical Malware Analysis, First Edition, No Starch Press, 2012.

Online Resources

- https://books.google.co.in/books/about/Reversing.html?id=G6M7DwAAQBAJ&redir_esc=y
- https://github.com/HACKE-RC/awesome-reversing?tab=readme-ov-file
- https://www.oreilly.com/library/view/reversing-secrets-of/9780764574818/pr01.html

SANDIP UNIVERSITY

School of Engineering and Technology

Department of English and Languages

Year: Third Year **Semester – V**

Course: Essential Aptitude Skills **Course Code: NHSA13**

Teaching Scheme (Hr. /Week)				Continuous Internal Assessment (CIA)					End Semester Examination		Total
L	T	P	C	CIA-1	CIA-2	CIA-3	CIA-4	Total	Theory	Practical	
-	-	4	2	10	20	10	10	50	-	50	100

Objectives	
1	To develop problem-solving skills by mastering quantitative aptitude concepts essential for applications.
2	To enhance logical and analytical thinking through structured reasoning techniques and critical thinking exercises.
3	To improve spatial and visual reasoning abilities by understanding 2D and 3D visualizations, patterns, and spatial problems.
4	To strengthen verbal reasoning and language proficiency for better comprehension, sentence structuring, and logical deductions.
5	To apply aptitude concepts in real-world scenarios, improving decision-making and problem-solving efficiency.

Unit Number	Details	Hours
1	**Unit 1: Quantitative Aptitude** • Understanding the Importance of Quantitative Aptitude • Fundamentals of Arithmetic and Algebra • Techniques for Solving Complex Numerical Problems • Application of Quantitative Aptitude	6
2	**Unit 2: Logical & Analytical Reasoning** • Understanding Logical and Analytical Thinking • Fundamentals of Deductive and Inductive Reasoning • Techniques for Solving Puzzles and Critical Thinking Questions • Application of Logical Reasoning in Real-World Problems	6

Document Reference	Revision No. / Date	Prepared By	Approved By
SUN/SOET /DOEL/2025	R5/31 May 2025	Dr. Sharayu Sonawane	Dr Mohini Gurav

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3	**Unit 3: Spatial & Visual Reasoning** • Understanding Spatial Intelligence in Problem-Solving • Fundamentals of 2D and 3D Visualizations • Techniques for Mirror Images, Patterns, and Arrangements • Application of Spatial Reasoning	6
4	**Unit 4: Verbal Reasoning and Logical Thinking** • Understanding Syllogisms, Analogies, and Logical Deductions • Para Jumbles and Sentence Rearrangement • Statement-Conclusion, Assumption-Based Questions • Error Detection, Cloze Tests, and Sentence Correction	6
5	**Unit 5: Aptitude Applications** • Understanding Aptitude in Decision-Making • Fundamentals of Speed, Distance, Probability, and Work-Time Problems • Techniques for Solving Data Sufficiency and Estimation Problems • Application of Mathematics in Real-World Scenarios	6
	Total	30

Course Outcomes: At the end of the course, students/learners will be able to:

CO 1	Develop their proficiency in quantitative aptitude by solving arithmetic, algebraic, and numerical problems efficiently.
CO 2	Demonstrate logical and analytical reasoning skills by solving puzzles, critical reasoning questions, and decision-making problems.
CO 3	Apply spatial and visual reasoning techniques to interpret patterns, 2D/3D figures, and visual problems.
CO 4	Enhance their verbal reasoning abilities by analyzing and solving syllogisms, statement-based questions, and sentence corrections.
CO 5	Utilize aptitude concepts such as speed-time-distance, probability, and data sufficiency in real-world problem-solving.

Resources	
Recommended Books	• R.S. Aggarwal. *Quantitative Aptitude for Competitive Examinations*. S. Chand Publishing, 2017. • Arun Sharma. *How to Prepare for Quantitative Aptitude for the CAT*. McGraw Hill Education, 2021. • R.S. Aggarwal. *A Modern Approach to Verbal and Non-Verbal Reasoning*. S. Chand Publishing, 2021. • Norman Lewis. *Word Power Made Easy*. Anchor Books, 2014. • Goleman, Daniel. *Emotional Intelligence: Why It Can Matter More Than IQ*. Bantam, 2006. • Nishit K. Sinha. *Logical Reasoning and Data Interpretation for the CAT*. Pearson Education India, 2022. • Pearson. *Objective General English*. Pearson India Education Services, 2022. • Tyra, Kundan, and Rishikant Tyra. *Quick Arithmetic: Shortcut Methods and Tricks for Competitive Exams*. BSC Publishing, 2019. • Rajesh Verma. *Fast Track Objective Arithmetic*. Arihant Publications, 2023. • Khattar, Dinesh. *The Pearson Guide to Quantitative Aptitude for Competitive Examinations*. Pearson Education, 2020. • Thorpe, Edgar. *Test of Reasoning for All Competitive Examinations*. Pearson Education India, 2017. • Sharma, A. K. *A Modern Approach to Logical Reasoning*. S. Chand Publishing, 2022.
Reference Books	• Barron's. *GRE: Graduate Record Examination*. Kaplan Test Prep, 2023. *(Also includes comprehensive reasoning and verbal aptitude content)* • Kaplan. *GRE Verbal Workbook*. Kaplan Publishing, 2023. • ETS. *The Official Guide to the GRE General Test*. Educational Testing Service, 2020. • Sharma, Ajay. *Magical Book on Quicker Maths*. BSC Publishing, 2020.
Web References	• **IndiaBix** • **GeeksforGeeks – Aptitude Section** • **Testbook** • **Gradeup (now BYJU'S Exam Prep)** • **Unacademy** • **Cambridge English**

Document Reference	Revision No. / Date	Prepared By	Approved By
SUN/SOET /DOEL/2025	R5/31 May 2025	Dr. Sharayu Sonawane	Dr Mohini Gurav

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SANDIP UNIVERSITY

School of Engineering and Technology

Department of English and Languages

CIA 1: Home Assignment for 10 Marks

Instructions for the Subject Teacher
The subject teacher has to create one assignment on each unit of knowledge level 3 as per Bloom's taxonomy.
Each assignment (Unit) carries 10 marks. Five assignments carry 50 marks. Convert 50 marks into 10 marks.
The student has to check the assignment from the subject teacher and upload a PDF of the same on Google Classroom before the deadline.
The subject teacher has to award the marks as per the rubrics of evaluation.
Note: The subject teacher may utilize the following CIA 1 as it is or as a reference to develop new CIA 1 according to the specific needs of the students.
Five assignments on each UNIT as per the syllabus and academic regulations

Unit	CO Mapping CO1/CO2/CO3/CO4/CO 5	Bloom's Taxonomy Level K1,K2,K3 etc	Q No	Assignment Name	ExaminationCategory	Date of Assignment Issued	Last Date of Submission of Assignment
	CO-1	L3	1	Question: If the sum of the first 50 natural numbers is S, what is the value of S?	CAT (Common Admission Test) – 2019		
	CO-1	L4	2	Question: A firm sells its product at ₹60 per unit. The total cost of production is ₹100, and the firm earns a total profit of ₹500. Later, the total cost increases by 30%. By what percentage should the price be increased to maintain the same profit level?	GATE (Graduate Aptitude Test in Engineering) – 2020		
	CO-1	L3	3	Question: The average of the monthly salaries of M, N, and S is ₹4000. The average of the monthly salaries of N, S, and P is ₹5000. The monthly salary of P is ₹6000. What is the monthly salary of M as a percentage of the monthly salary of P?	SSC (Staff Selection Commission) – CGL 2018		
	CO-1	L3	4	Question: A car travels 8 km in the first quarter of an hour, 6 km in the second quarter, and 16 km in the third quarter. What is the average speed of the car?	UPSC (Civil Services Examination) – CSAT 2017		

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SUN/SOET /DOEL/2025	R5/31 May 2025	Dr. Sharayu Sonawane	Dr Mohini Gurav

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	CO-1	L2	5	Question: If p:q = 1:2, q:r = 4:3, and r:s = 4:5, what is the ratio p:s?	MPSC (Maharashtra Public Service Commission) – 2019		
2	CO-2	L4	1	Question: All cats are dogs. Some dogs are rats. Which of the following conclusions logically follow? A) Some cats are rats. B) All dogs are cats. C) Some rats are cats. D) None of the above	CAT (Common Admission Test) – 2020		
	CO-2	L3	2	Question: If 'A' is taller than 'B' and 'B' is taller than 'C', which of the following statements is true? A) A is the tallest. B) C is taller than A. C) B is the tallest. D) None of the above.	GATE (Graduate Aptitude Test in Engineering) – 2018		
	CO-2	L4	3	Question: In a certain code language, 'APPLE' is written as 'EQQPI'. How is 'ORANGE' written in that code?	SSC (Staff Selection Commission) – CGL 2019		
	CO-2	L5	4	Question: If it is raining, then the ground is wet. The ground is wet. Can we conclude that it is raining?	UPSC (Civil Services Examination) – CSAT 2016		
	CO-2	L5	5	Question: A statement is followed by two assumptions: Statement: All students must submit their assignments by Monday. Assumption I: Students are capable of submitting assignments by Monday. Assumption II: Assignments submitted after Monday will not be accepted. Which of the assumptions is/are implicit?	MPSC (Maharashtra Public Service Commission) – 2020		
3	CO-3	L3	1	Question: A cube is painted on all sides and then cut into smaller cubes of equal size. How many smaller cubes will have exactly two sides painted?	SSC CGL (Staff Selection Commission Combined Graduate Level) – 2017		
	CO-3	L4		Question: A square sheet of paper is folded along its diagonal to form a	UPSC CSAT (Civil Services		

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			2	triangle. This triangle is then folded along its height to form a smaller triangle. What is the shape of the base of the final folded structure?	Aptitude Test) 2004		
	CO-3	L5	3	Question: Consider a square sheet of side 1 unit. The sheet is first folded along the main diagonal. This is followed by a fold along its line of symmetry. The resulting folded shape is again folded along its line of symmetry. What is the area of each face of the final folded shape, in square units?	GATE (Graduate Aptitude Test in Engineering) 2012		
	CO-3	L4	4	Question: Which of the following 3D shapes can be formed by folding the given 2D net? (Options provided with various 3D shapes)	CAT (Common Admission Test) 2000		
	CO-3	L3	5	Question: A dice has numbers 1 to 6 on its faces. If the dice is rolled and the face with the number 3 is on top, which number will be on the bottom face?	MPSC (Maharashtra Public Service Commission) – 2019		
	CO-4	L4	1	Question: Statements: All pens are books. All books are papers. Some papers are notebooks. No notebook is a pen.Conclusions: Some pens are papers. No pen is a notebook. Some books are notebooks. Some notebooks are not pens. Which of the conclusions logically follow?	SSC CGL (Staff Selection Commission Combined Graduate Level) – 2018		
	CO-4	L3	2	Question: Find the analogy: Doctor is to Hospital as Teacher is to: A) Class B) School C) Student D) Education	UPSC Civil Services Examination – 2017		

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	CO-4	L5	3	Question: Rearrange the following sentences to form a coherent paragraph: A) The Industrial Revolution brought about significant changes. B) This led to urbanization and changes in lifestyle. C) Many people moved to cities for work. D) Factories became the centers of production.	ICAT (Common Admission Test) – 2020		
	CO-4	L4	4	Question: Statement: "All that glitters is not gold." Conclusion: A) All gold items glitter. B) Some non-gold items may glitter. Which of the conclusions logically follow?	MPSC (Maharashtra Public Service Commission) – 2019		
	CO-4	L3	5	Question: Identify the grammatical error in the following sentence: "Each of the boys are playing football."	SSC CGL (Staff Selection Commission Combined Graduate Level) – 2018		
	CO4	L3	2	Question: Fill in the blanks: "The committee has decided to _____ the meeting until next week due to unforeseen circumstances." Options: A) postpone B) prepone C) cancel D) advance	IBPS PO (Institute of Banking Personnel Selection Probationary Officer) – 2019		
	CO4	L3	2	Question: Identify the correct sentence: A) She gave me an advice. B) She gave me a advice. C) She gave me some advice. D) She gave me many advices.	CAT (Common Admission Test) – 2019		
	CO-5	L3	1	Question: A train 150 meters long is running at 54 km/h. In how much time will it cross a platform 150 meters long? Options: A) 10 sec B) 15 sec C) 20 sec D) 25 sec	SSC CGL 2020		

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	CO-5	L4	2	Question: A box contains 5 red balls and 3 blue balls. Two balls are drawn randomly. What is the probability that both are red? Options: A) 10/21 B) 5/21 C) 1/2 D) 2/7	GATE 2019 (General Aptitude)		
	CO-5	L4	3	Question: A and B together can complete the work in 12 days. A alone can complete it in 20 days. How long will B alone take?	SSC JE 2018		
	CO-5	L5	4	Question: Is $x > 5$? $x + 2 > 7$ $x > 4$	CAT 2021		
	CO5	L3	5	Question: Estimate the value of √(123^2 + 5^2) Options: A) ~123 B) ~128 C) ~130 D) ~125	GATE 2020		
	CO5	L2	6	Question: An engineer needs to design a water tank to be filled in 6 hours. If the flow rate is 1500 liters/hour, what should be the capacity of the tank?	GATE (General Aptitude) – 2022		
	CO5	L5	7	Question: A product's price increased by 20% and then decreased by 25%. What is the net percentage change?	CAT 2019		
	CO5	L6	8	Question: An engineer has to choose between two machines: Machine A costs ₹1,00,000 and lasts 10 years. Machine B costs ₹60,000 and lasts 5 years. Which is more economical, assuming no running cost?	UPSC Engineering Services (IES) – 2017		

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CIA 1 Rubrics: Home Assignment for 10 Marks

Category	Marks	Excellent (100%)	Very Good (75%)	Good (50%)	Fair (25%)	Poor (0%)
Vocabulary and Grammar	2	Vocabulary and grammar are clear and greatly add to the reader's understanding of the procedures	Vocabulary and grammar are clear and greatly add to the reader's understanding of the procedures	Vocabulary and grammar are somewhat difficult to understand	Many errors in vocabulary and grammar choice that are severely difficult to understand or are not used	Not Submitted
		2	1.5	1	0.5	0
Completion	2	All questions are completed.	75% questions are completed	50% questions are completed	25% questions are completed	Not Submitted
		2	1.5	1	0.5	0
Punctuality	3	Submission of the assignment before the last date	Submission of the assignment on the last date	Submission of the assignment within one week	Submission of the assignment within two weeks	Not Submitted
		3	2.25	1.5	0.75	0
Neatness and Organization	3	The work is presented in a neat, clear, organized fashion that is easy to read.	The work is presented in a neat and organized fashion that is usually easy to read.	The work is presented in an organized fashion, but may be hard to read at times.	The work appears sloppy and unorganized. It is hard to know what information goes together.	Not Submitted
		3	2.25	1.5	0.75	0

CIA 2 for 20 Marks (MCQ-based Written Exam on Each Unit)

Document Reference	Revision No. / Date	Prepared By	Approved By
SUN/SOET /DOEL/2025	R5/31 May 2025	Dr. Sharayu Sonawane	Dr Mohini Gurav

SANDIP UNIVERSITY MAHARASHTRA NASHIK

REGISTRAR
SANDIP UNIVERSITY
NASHIK

SANDIP UNIVERSITY

School of Engineering and Technology

Department of English and Languages

CIA 3 Open Source Presentation for 10 Marks
Approximately 10 to 12 Slides
Rubrics for (CIA 3)

Category	Marks	Excellent	Very Good	Good	Fair	Poor
Content	**4**	Content is thorough, accurate, and highly relevant. Demonstrates deep understanding and insight.	Content is accurate and relevant. Shows a good understandin g of the topic.	Content is mostly accurate and relevant. Some areas lack depth or clarity.	Content is incomplete or inaccurate. Demonstrates limited understanding.	Content is entirely irrelevant.
		4	3	2	1	0
Delivery	**3**	Engages audience effectively, using varied and compelling delivery methods. Uses positive body language throughout.	Engages the audience well, with clear delivery. Uses positive body language throughout.	Engages the audience adequately but may lack confidence or enthusiasm. Occasionally breaks eye contact, and body language appears unsure.	Engages the audience inconsistently, with unclear delivery. Rarely maintains eye contact or uses appropriate body language.	Fails to engage the audience effectively. Does not maintain eye contact or use body language effectively.
		3	2	1.5	1	0
Presentation	**3**	The Presenter was familiar with the material and did not from the slides or rely on notes	The presentation was well rehearsed, and the presenter spoke clearly	The presenter showed enthusiasm for the subject matter	The presenter made eye contact with the audience	Many requirements are not met
		3	2	1.5	1	0

CIA 4 - Attendance and Behaviour (10 Marks)
MCQ- Based End Semester Examination (50 Marks)

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SUN/SOET /DOEL/2025	R5/31 May 2025	Dr. Sharayu Sonawane	Dr Mohini Gurav

SANDIP UNIVERSITY MAHARAVALI NASHIK

REGISTRAR
SANDIP UNIVERSITY
NASHIK

SANDIP UNIVERSITY, NASHIK, MS (INDIA)
SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
Department of Computer Science and Engineering
MAHIRAVANI, TRIMBAK ROAD, TAL & DIST: NASHIK-422213, MAHARASHTRA,INDIA

Year: Third Year **Semester – V**
Course: Tableau and Power BI **Course Code: VAC**

Teaching Scheme (Hrs. /Week)				Continuous Internal Assessment (CIA)					End Semester Examination		Total
L	T	P	C	CIA-1	CIA-2	CIA-3	CIA-4	Lab	Theory	Lab	
3	0	0	3	10	20	10	10	--	50	--	100
Max. Time, End Semester Exam (Theory) - 3Hrs.									End Semester Exam (Lab) - 0Hrs.		

Prerequisites:
basic database concepts and data handling, basic programming or analytical thinking skills.

Objectives:	
Students are able to:-	
1	To understand how to connect, clean, transform, and model data using Power BI for effective data visualization.
2	To develop the ability to create dynamic and interactive reports using charts, maps, slicers, filters, and drill-down features in Power BI.
3	To learn how to publish, share, and analyze dashboards using Power BI Service for business analytics applications.
4	To understand the use of Tableau for connecting data sources and creating interactive visualizations and dashboards.
5	To develop the skills to create KPIs, perform analytical calculations, and design interactive dashboards for data-driven decision making.

Unit No	Details	Hours
1	**UNIT I : Data Visualization using Power BI** Connect to multiple data sources like Excel, CSV, and SQL with ease, Clean, transform, and model data using Power Query, ,	9
2	**UNIT II : Building Interactive Reports in Power BI** Create dynamic visualizations – charts, maps, and dashboards, Build interactive reports with slicers, filters, drill-downs, and tooltips	9
3	**UNIT III: Power BI Service and Business Analytics Applications** Publish and share dashboards securely via Power BI Service, Gain insights from data using visual storytelling techniques, Learn real-world business use cases: sales, finance, HR analytics	9
4	**UNIT IV : Introduction to Tableau and Data Visualization** Dive into data by connecting Tableau with files, databases, or cloud sources effortlessly, Explore and shape raw data visually with drag-and-drop simplicity,	9

	Transform numbers into stunning visuals like heatmaps, treemaps, and interactive maps, Build dashboards that respond to clicks, filters, and user selections in real time.	
5	**UNIT V : Creating KPIs and Interactive Data Dashboards** Create powerful calculations, rankings, and KPIs without writing complex code, Tell compelling data stories that highlight patterns, trends, and outliers, Publish interactive dashboards online to share insights with teams or clients	9

Outcomes:	
On completion of the course, learner will be able to :-	
CO1:	Apply Power BI to connect, clean, transform, and model data from multiple data sources for visualization.
CO2:	Create interactive reports and dashboards using charts, maps, slicers, filters, and drill-down features in Power BI.
CO3:	Utilize Power BI Service to publish, share dashboards, and analyze business data using visual storytelling techniques.
CO4:	Use Tableau to connect data sources and create interactive visualizations and dashboards for data exploration.
CO5:	Develop KPIs, analytical calculations, and interactive dashboards to support data-driven decision making.

Text Books

1. D. Knight, B. Lachev, and M. Lorenz, Microsoft Power BI Quick Start Guide, 2nd ed. Birmingham, UK: Packt Publishing, 2021.
2. J. Stirrup, A. Nandeshwar, A. Ohmann, and M. Floyd, Tableau: Creating Interactive Data Visualizations, Birmingham, UK: Packt Publishing, 2015.
3. J. Dougherty and I. Ilyankou, Hands-On Data Visualization: Interactive Storytelling from Spreadsheets to Code, Sebastopol, CA, USA: O'Reilly Media, 2021.

Reference Books

1. A. Kolokolov and M. Zelensky, *Data Visualization with Microsoft Power BI*, Sebastopol, CA, USA: O'Reilly Media, 2024.
2. A. Loth, *Visual Analytics with Tableau*, Hoboken, NJ, USA: John Wiley & Sons, 2019.

Online Resources

- **Power BI Community Forum**, https://community.powerbi.com
- **Tableau Public Gallery**, https://public.tableau.com
- **Tableau Learning Resources**, https://www.tableau.com/learn